

# Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

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## Abstract

Large language models (LMs) store vast factual knowledge, yet turning it into a reliable, *set-valued* knowledge base, where a subject may have zero, one, or many objects, remains hard. The LM-KBC 2026 shared task poses this in a strict *closed-book* setting: an open-weight model of at most 32B parameters, with no retrieval, no external knowledge, and no fine-tuning. We start from the metric and observe that it rewards *abstention*: since an empty prediction scores a perfect  $F_1$  against an empty gold set, predicting nothing everywhere already reaches 0.175 relation-averaged macro- $F_1$ , near the official baseline’s 0.273. We therefore treat *when to answer* as a first-class decision and build a per-relation pipeline that grounds each prompt in the relation’s exact scope, decides abstention from the sampled set rather than by instruction, and takes a median over samples for numeric relations. Our **Gemma-4-31B** system (30.7B, the strongest in-budget model we could access) scores 0.459/0.490 (relation-averaged/micro) from a single sample, beating the official baseline on all six relations, while a smaller 4-bit Qwen2.5-14B, even with self-consistency voting, reaches only 0.416: in this comparison a larger single-sample model outcores a smaller voted one. Generation is cached, so decoding reproduces without a GPU. We release our code.<sup>1</sup>

## 1 Introduction

Pre-trained language models (LMs) store a remarkable amount of factual knowledge in their parameters (Petroni et al., 2019; Roberts et al., 2020), and a body of work probes and extracts it (Jiang et al., 2020; Zhong et al., 2021; AlKhamissi et al., 2022). This raises the prospect of building a structured knowledge base (KB) directly from an LM rather than from curated sources (Dong et al.,

2014; Vrandečić and Krötzsch, 2014). Doing so is harder than answering a single factual question: a KB entry is a *complete set* of facts, so the model must decide not only what is true but how many objects, possibly none, a subject and relation admit.

The LM-KBC 2026 shared task (Alivanistos et al., 2022; Kalo et al., 2023, 2025) formalises this: given a subject  $s$  and relation  $r$ , predict the complete object set  $\{o_1, \dots, o_k\}$ , where  $k$  may be zero, one, or many (Section 3). Predictions are scored by macro precision, recall, and  $F_1$ , with alias matching for entities and a 5% tolerance for numeric relations.

This is not single-answer probing. Unlike benchmarks such as LAMA (Petroni et al., 2019), which assume one gold object and reward ranking it highly, the task demands the *entire* set, and the empty set is a legitimate, frequent answer: a living person has no recorded city of death, and an island country has no land border. LM outputs are also stochastic, relations range from a single number to hundreds of recipients, and the model must recognise when it does not know an answer and *abstain* rather than fabricate one.

The 2026 edition is strictly *closed-book*: any open-weight model up to 32B parameters, but no web search, retrieval, external corpus, or fine-tuning. This forecloses the retrieval and tuning that strong entries in earlier editions relied on (Zhang et al., 2023; Asai et al., 2024), leaving parametric recall and prompt design as the only levers.

We start from the evaluation metric rather than the model. Because macro- $F_1$  scores an empty prediction against an empty gold set as perfect, predicting nothing everywhere already attains 0.175 relation-averaged macro- $F_1$ , competitive with the official baseline (0.273) and higher on the most null-heavy relations (Section 3). Points therefore come not from answering everything, but from recalling the few numeric values and, above all,

<sup>1</sup><https://github.com/ANON/lm-kbc-2026>

from knowing when *not* to answer. We make abstention a first-class decision: for each null-heavy relation we sample several times and keep an object only when the samples agree, collapsing to  $\emptyset$  when the model is inconsistent, while scope-grounded prompts and median aggregation handle the rest (Figure 1).

This is a shared-task system paper making three claims of decreasing generality. First, a metric observation (Section 3): since a correct empty set scores a perfect  $F_1$ , the empty baseline already rivals the official one, so the task rewards *when to answer* as much as what to answer. This follows from the metric but is easy to miss, and it shapes our design. Second, a simple, reproducible pipeline (Section 4) with no trainable parameters, combining standard parts: scope-grounded prompting, self-consistency voting for abstention (a fixed a-priori  $\theta=0.5$ ), and median aggregation for numerics; at the deployed  $\theta=0.5$  the voting lifts a weak 0.5B model from 0.173 to 0.363 on the null-heavy relations. Third, and most useful, an empirical finding: in our one controlled comparison at the  $\leq 32B$  budget, a single-sample Gemma-4-31B (0.459/0.490, beating the official baseline on all six relations) outscores the smaller 4-bit Qwen2.5-14B even with self-consistency voting (0.416); whether voting also helps the 30B is untested, as API limits kept it off (Section 7).

## 2 Background and Related Work

Pretrained language models have long been probed as implicit knowledge bases, a line that Petroni et al. (2019) opened by casting factual recall as cloze probing. Subsequent work sharpened the prompts (Jiang et al., 2020; Shin et al., 2020; Zhong et al., 2021), quantified how much knowledge a model’s parameters hold (Roberts et al., 2020), and questioned how reliably that knowledge can be read out (Cao et al., 2021; AlKhamissi et al., 2022). These probes share a common assumption: a single gold object per query, scored by how highly the model ranks it. We instead target the set-valued, alias-graded variant in which the empty set is a valid answer, a setting far closer to materialising an actual knowledge base than to ranking a single token (Pan et al., 2023).

A parallel thread turns language models toward building knowledge bases rather than merely probing them, and the LM-KBC shared task is its clearest instance. Across its editions (Alivanis-

tos et al., 2022; Kalo et al., 2023, 2024, 2025) the task has moved from masked-LM probing and prompt ensembling toward instruction-tuned generative systems, with some earlier tracks permitting retrieval or fine-tuning (Zhang et al., 2023). The 2026 edition removes those options with its closed-book, no-tuning,  $\leq 32B$  rules, and it is exactly this stripped-down regime that forces the parametric-recall-with-abstention design we pursue.

The usual route to knowledge-intensive accuracy under such pressure is retrieval augmentation (Lewis et al., 2020; Guu et al., 2020), which helps most precisely when parametric memory is weak (Mallen et al., 2023). Self-RAG (Asai et al., 2024) pushes this further by letting the model itself decide *when* to retrieve. We borrow the spirit of a model-driven, adaptive decision, but because retrieval is forbidden we redirect it to a different question: not what to fetch, but *when to answer at all*.

That decision draws together three further threads that converge in our method. Self-consistency (Wang et al., 2023; Chen et al., 2023) samples a model repeatedly and aggregates the generations to select a single answer; abstaining when uncertain is the classic reject option (Geifman and El-Yaniv, 2017), studied in NLP as calibration (Guo et al., 2017), unanswerable QA (Rajpurkar et al., 2018), and selective QA (Cole et al., 2023), and language models retain some awareness of what they know (Kadavath et al., 2022). We combine these: cross-sample *disagreement* (Kuhn et al., 2023; Manakul et al., 2023) serves as our zero-parameter confidence signal, and, unlike majority-vote self-consistency, which always emits an answer, we let that signal trigger *abstention*. Our setting differs from all of the above in one decisive way, and it is what makes abstention necessary rather than optional: the objects form a set of unbounded cardinality, the prediction is closed-book and untuned, and the empty set is itself a valid, high-value prediction rather than a failure to be avoided.

## 3 Task and Metric Analysis

The dataset comprises six Wikidata relations (Table 1) that fall into three groups with sharply different answer shapes. The *numeric* relations HASCAPACITY and HASAREA have a single scalar object; the *large-set* relation AWARD-

| Relation                     | #val | empty% |
|------------------------------|------|--------|
| COUNTRYLANDBORDERSCOUNTRY    | 68   | 25     |
| PERSONHASCITYOFDEATH         | 100  | 45     |
| COMPANYTRADESATSTOCKEXCHANGE | 100  | 35     |
| AWARDWONBY                   | 10   | 0      |
| HASCAPACITY                  | 100  | 0      |
| HASAREA                      | 100  | 0      |

Table 1: Validation statistics. AWARDWONBY averages  $\sim 148$  objects per subject (max 641); three string relations are often legitimately empty.

WONBY has up to hundreds of recipients per subject; and three *null-heavy* string relations, COUNTRYLANDBORDERSCOUNTRY, PERSONHASCITYOFDEATH, and COMPANYTRADESATSTOCKEXCHANGE, leave a large fraction of subjects with no valid object at all. Each relation carries a precise scope (land-only borders, city-granular death location, the highest published capacity), which we copy verbatim into the prompt so that the model targets the same notion the gold set was constructed under.

Predictions are scored against the gold object set by macro precision and recall. A predicted string counts as a true positive if its normalised form (lower-cased, with diacritics and punctuation removed) matches any alias of a gold object, where each gold object is given as a set of surface aliases; a numeric prediction counts if it lies within a 5% relative tolerance of the gold value. Two boundary conventions drive everything that follows: precision is defined as 1 when the prediction is empty, and recall as 1 when the gold set is empty. We report *relation-averaged* macro- $F_1$  (the mean of the six per-relation scores) and *micro-averaged*  $F_1$  (over all instances).

These conventions follow directly from the metric definition, but their effect is easy to underestimate: because an empty prediction against an empty gold set is a perfect  $F_1$ , predicting the empty set for *every* validation instance already scores the “Empty” bars in Figure 3, averaging 0.175 relation-averaged macro- $F_1$  (0.203 micro). This nearly matches the official baseline’s 0.273 relation-averaged macro- $F_1$  and *beats* it on the two most null-heavy relations (PERSONHASCITYOFDEATH 0.45 vs. 0.16; COMPANYTRADESATSTOCKEXCHANGE 0.35 vs. 0.32). The metric therefore rewards two narrow abilities rather than answering everything: recalling the numeric values, and abstaining precisely when no object exists. Our pipeline pulls

#### Algorithm 1 CALIBRATEDABSTENTION

**Require:** samples  $S=\{s_1, \dots, s_N\}$ , threshold  $\theta$ , *multi*  
1:  $c[o] \leftarrow \#\{s_i : o \in \text{PARSE}(s_i)\}$  for each object  $o$   
2: **if** *multi* **then return**  $\{o : c[o]/N \geq \theta\}$   
3: **else**  
4:  $o^* \leftarrow \arg \max_o c[o]$ ; **return**  $c[o^*]/N \geq \theta ? \{o^*\} : \emptyset$   
5: **end if**

exactly these two levers.

## 4 System

**Overview.** The pipeline (Figure 1) maps each subject–relation pair  $(s, r)$  to one chat prompt, decodes a cached set of samples ( $N$  for voting,  $n$  for the numeric median), and aggregates them per relation kind into an object set. It adds no trainable parameters of its own and targets the two analysis levers: abstention and numeric recall.

**Scope-grounded prompts.** Each relation’s system message states the task, the *exact* ground-truth scope (copied verbatim from the official definitions), and the required answer form (Table 2); the model emits one JSON object  $\{\text{"answers"} : [\dots]\}$  (Figure 2), which makes the empty case unambiguous. Few-shot exemplars come from the training split; for null-heavy relations we interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed. Even a 0.5B model emits valid JSON on 100% of instances, so format is never a bottleneck.

**Calibrated abstention.** Scope grounding tells the model *what* counts as an answer, but not *when* to stay silent. A bare instruction to “answer empty when unsure” is unreliable: left unconstrained, a small model names a death-city for living people and invents stock exchanges, dropping precision *below* the empty baseline (Section 7). Instead, for the three null-heavy relations we sample  $N=3$  completions at temperature  $\tau=0.7$  and keep an object only if it appears in at least a fraction  $\theta$  of samples; single-object relations abstain entirely if even the top answer is below  $\theta$  (Algorithm 1). When the model guesses, samples disagree and the prediction collapses to  $\emptyset$ , exactly where the metric rewards silence. We disable voting for AWARDWONBY (recall-bound) and the numeric relations.

**Numeric aggregation.** The numeric relations need a different aggregator: HASCAPACITY and HASAREA take a single number graded within 5%. We sample  $n=5$  completions and take the *median*

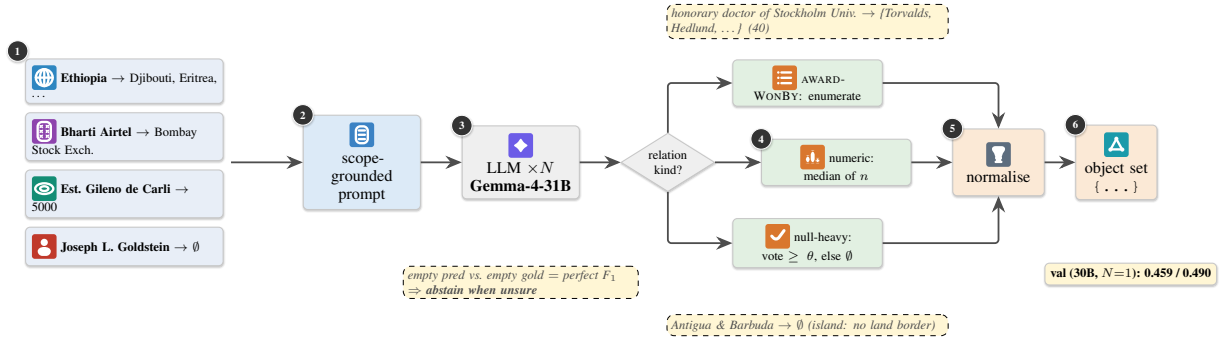


Figure 1: System pipeline (no trainable parameters). Real dataset instances (left, stage 1), shown with entity-type icons and their gold objects, are each rendered into a scope-grounded chat prompt (stage 2); the model is sampled  $N$  times (stage 3, cached), and the samples are aggregated by relation kind (stage 4): *median* for numeric relations, consistency *voting with abstention* (keep an object only if it appears in  $\geq \theta$  of samples, else  $\emptyset$ ) for the null-heavy string relations, and *enumeration* for AWARDWONBY; a final normalisation matches the evaluator exactly. Dashed boxes are real worked examples: the island country ANTIGUA & BARBUDA correctly collapses to  $\emptyset$ , as does the still-living scientist JOSEPH L. GOLDSTEIN (no city of death). The design follows from the metric insight (lower-left badge). The reported Gemma-4-31B score uses a single sample ( $N=1$ , voting off); the  $N$ -sample voting path is exercised by the 14B ablation (Section 7).

| Relation                     | Scope stated in the prompt                                                            | Answer form      |
|------------------------------|---------------------------------------------------------------------------------------|------------------|
| COUNTRYLANDBORDERSCOUNTRY    | countries sharing a <i>land</i> border; maritime borders excluded; an island has none | countries, or [] |
| PERSONHASCITYOFDEATH         | the <i>city</i> where the subject died; empty if alive or the city is unknown         | one city, or []  |
| COMPANYTRADESATSTOCKEXCHANGE | stock exchange(s) on which the company is publicly listed; empty if unlisted          | exchanges, or [] |
| AWARDWONBY                   | every recipient of the award (often tens to hundreds)                                 | names (long)     |
| HASCAPACITY                  | maximum spectator capacity in people; highest published value                         | one integer      |
| HASAREA                      | total surface area in $\text{km}^2$ (land + inland water)                             | one number       |

Table 2: Per-relation prompt specification. Each system message states the relation’s exact scope (copied from the official definitions) and the required answer form; the model replies with a single JSON object {"answers": [...]}.

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SYS: precise KB engine (closed book).
TASK: city where the subject died.
SCOPE: city granularity; [] if
alive/unknown.
Reply only {"answers":[...]}.
U: Egbert Mulder
A: {"answers":["Groningen"]}
U: Dave Keon
A: {"answers":[]}
U: <query> A:

```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed exemplars teach abstention; output is forced JSON.

relations (handling commas, units, scale words), and de-duplicates. Normalisation (lower-casing, diacritic/punctuation removal) is byte-identical to the official evaluator; pushing gold answers back through the parser scores 1.000, confirming it is lossless.

## 5 Experimental Setup

**Models.** Our primary system uses **Gemma-4-31B-it** (Gemma Team, 2026) (Gemma-4-31B for brevity), a 30.7B-parameter open-weight model within the 32B cap. It exceeds the memory of our single 16 GB GPU, so we serve it through a hosted OpenAI-compatible API. The weights are open and we apply no fine-tuning; the end-point serves the unmodified weights as a plain text-completion service (no retrieval, tools, or augmentation), so it is compute-only and the closed-book rules hold. For the self-consistency ablations



(Section 7) we additionally run **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B) locally in 4-bit, together with a tiny **Qwen2.5-0.5B-Instruct** used to probe abstention on a poorly-calibrated model. Several nominally “32B” models in fact exceed the cap (Qwen2.5-32B at 32.5B, Qwen3-32B at 32.8B) and are ineligible.

**Decoding.** At decode time, the pipeline selects an aggregator per relation kind (Section 4): the median over  $n=5$  samples for numeric relations, consistency voting over  $N=3$  samples at threshold  $\theta=0.5$  for null-heavy relations (Algorithm 1), and a single greedy enumeration for AWARDWONBY. We evaluate these self-consistency configurations (temperature  $\tau=0.7$ ) on Qwen2.5-14B, which we can sample freely. API rate limits preclude multi-sampling Gemma-4-31B at scale, so we decode it with a *single* sample per query and confine the self-consistency ablations to the 14B (Section 6). Prompts are 8-shot for every relation except AWARDWONBY (2-shot), with exemplars from the training split (Brown et al., 2020).

**Hardware.** Qwen2.5-14B runs on a single NVIDIA T4 (16 GB) via HuggingFace Transformers in 4-bit (bitsandbytes nf4, float16 compute) (Dettmers et al., 2023); quantisation only fits the model to the hardware and does not change the counted parameter budget. Gemma-4-31B is queried over its API from the same machine. Each model processes the full 478-instance validation set.

**Baselines.** We compare against two references on the same validation split: the *empty* baseline, which predicts  $\emptyset$  for every instance (Section 3), and the *official* organiser baseline, a few-shot prompting system whose per-relation scores we take from the task repository.

**Reproducibility.** The pipeline caches every raw sample, so all post-processing (parsing, consistency voting, median aggregation, and the threshold sweep) is a deterministic, GPU-free function of that cache. Every ablation in Section 7 therefore comes from a single generation pass and reproduces without a GPU. We release all code and prompts.

## 6 Results

Our primary system, **Gemma-4-31B**, scores **0.459** relation-averaged macro- $F_1$  (**0.490** micro) on val-

| Relation           | Empty | Base  | 14B          | <b>30B</b>   |
|--------------------|-------|-------|--------------|--------------|
| countryLandBorders | 0.250 | 0.679 | 0.900        | <b>0.953</b> |
| companyTrades      | 0.350 | 0.320 | 0.616        | <b>0.722</b> |
| personCityOfDeath  | 0.450 | 0.160 | <b>0.480</b> | 0.460        |
| hasArea            | 0.000 | 0.310 | 0.330        | 0.330        |
| hasCapacity        | 0.000 | 0.100 | 0.090        | <b>0.170</b> |
| awardWonBy         | 0.000 | 0.069 | 0.081        | <b>0.120</b> |
| Avg. of relations  | 0.175 | 0.273 | 0.416        | <b>0.459</b> |
| Micro (all inst.)  | 0.203 | 0.284 | 0.447        | <b>0.490</b> |

Table 3: Validation macro- $F_1$  by model. “Empty” is the measured empty baseline (predict nothing everywhere), “Base” the official baseline; **30B** is our primary Gemma-4-31B system, 14B is Qwen2.5-14B. Best per row in bold. Gemma-4-31B beats the official baseline on *all six* relations.

idation, versus 0.175/0.203 for the empty baseline, 0.273/0.284 for the official baseline, and 0.416/0.447 for the smaller Qwen2.5-14B. Figure 3 and Table 3 give the per-relation picture. Gemma-4-31B *beats the official baseline on all six relations* and the 14B on four of six. Over the 14B, the 30B’s gains concentrate where the model already has usable knowledge: the biggest 14B  $\rightarrow$  30B jumps are on COMPANYYTRADESATSTOCKEXCHANGE (0.62  $\rightarrow$  0.72) and HASCAPACITY (0.09  $\rightarrow$  0.17, nearly doubled but still low in absolute terms), while the near-saturated COUNTRYLANDBORDERSCOUNTRY reaches 0.953. It ties the 14B on HASAREA and trails only on PERSONHASCITYOFDEATH (0.46 vs. 0.48), the sole relation where the single-sample 30B forgoes the abstention voting that the 14B uses, so the one loss is a deliberate consequence of the decoding budget rather than a knowledge gap. Strikingly, the 30B attains this overall lead *without* self-consistency or abstention, from a single sample. We could not run voting on the 30B, so we isolate the decode-time levers on the smaller models in Section 7.

## 7 Analysis

Because the headline 30B result is single-sample, we isolate the decode-time levers on the 14B and 0.5B models, which we can sample freely; all ablations use the same cached samples.

**Abstention.** Abstention rescues weak models and refines strong ones. Table 4 isolates it on the three null-heavy relations. On a poorly-calibrated 0.5B model the effect is dramatic (0.173  $\rightarrow$  0.363 at the deployed  $\theta=0.5$ , up to 0.372 under the

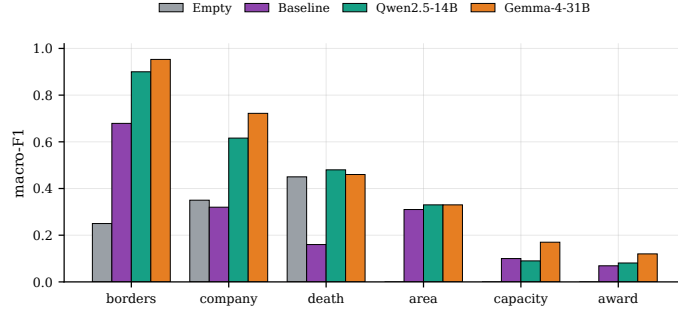


Figure 3: Per-relation validation macro-F<sub>1</sub>: empty baseline, official baseline, Qwen2.5-14B, and Gemma-4-31B. Gemma-4-31B beats the official baseline on all six relations.

|                            | Off   | $\theta=0.5$ | $\theta=0.67$ |
|----------------------------|-------|--------------|---------------|
| 0.5B (avg of 3 null-heavy) | 0.173 | 0.363        | <b>0.372</b>  |
| 14B (avg of 3 null-heavy)  | 0.652 | 0.665        | <b>0.666</b>  |

Table 4: Self-consistency abstention, relation-averaged over the three null-heavy relations only. “Off” decodes a single sample.

sweep): without voting it over-answers and falls *below* the empty baseline (0.350). The 14B is already above that baseline without voting (0.652), so voting adds less (0.652  $\rightarrow$  0.666) but still helps. Abstention buys precision: on PERSONHASCITY-OFDEATH it moves  $P=0.77 \rightarrow 0.81 \rightarrow 0.89$  as  $\theta$  rises, trading a little recall. It suppresses wrong guesses, not correct answers.

**Setting the threshold.** If abstention drives the string relations, the natural next question is how to set its threshold. Our deployed systems use a single fixed  $\theta=0.5$  (a simple majority of three samples); the sweep in Figure 4 (left) is diagnostic, and since it is run on the split we report on, its per-relation optima are optimistic rather than values we tune back in. Still, PERSONHASCITY-OFDEATH (45% empty) peaks at a *lower*  $\theta$  while the others tolerate aggressive voting, so a fixed majority leaves little on the table. Figure 4 (right) varies the number of numeric samples: median aggregation is non-monotonic on HASAREA and actually slightly *hurts* HASCAPACITY on the 14B (single sample 0.10 vs. median-5 0.09). Decode variance is not the bottleneck; the gap is knowledge-bound: the capacity guesses are the right order of magnitude but outside the 5% band (e.g. Spartan Stadium: predicted 37,000 vs. gold 75,005), and even for the 30B both numeric relations stay low (0.17, 0.33).

**The AWARDWONBY recall cap.** The relation AWARDWONBY, which we beat the baseline on only narrowly, is recall-capped, and this limit constrains both models. Gold sets average  $\sim 148$  recipients (max 641), but in the 14B ablation the model enumerates only  $\sim 11$  names; its outputs are *not* truncated, it closes the JSON early, capping recall near  $11/148 \approx 0.07$ . The listed names are often plausible but wrong ( $P=0.36$ ). Parametric recall and a reluctance to enumerate bound this relation; within a closed-book single model it is largely unsolved, and a larger in-budget model or multi-step decomposition would help most.

**Qualitative behaviour.** Table 5 shows representative Gemma-4-31B outputs: exact on well-known facts and near-complete on borders, with the two error modes above visible, a numeric near-miss on HASAREA and plausible-but-wrong famous names on AWARDWONBY.

## 8 Conclusion

We treated *when to answer* as the central decision in closed-book, set-valued KB construction, and derived each component from one metric fact: because macro-F<sub>1</sub> rewards a correct empty set, abstention pays off, numeric recall is scarce, and scope grounding curbs over- and under-prediction. Instantiated with Gemma-4-31B, the system reaches 0.459/0.490 (relation-averaged/micro) and beats the official baseline on all six relations, reproducibly from a cached pass. The durable lesson is comparative: a single-sample Gemma-4-31B outscores a smaller voted Qwen2.5-14B, while voting mainly rescues weaker models; pairing voting with the 30B is the clearest next step.

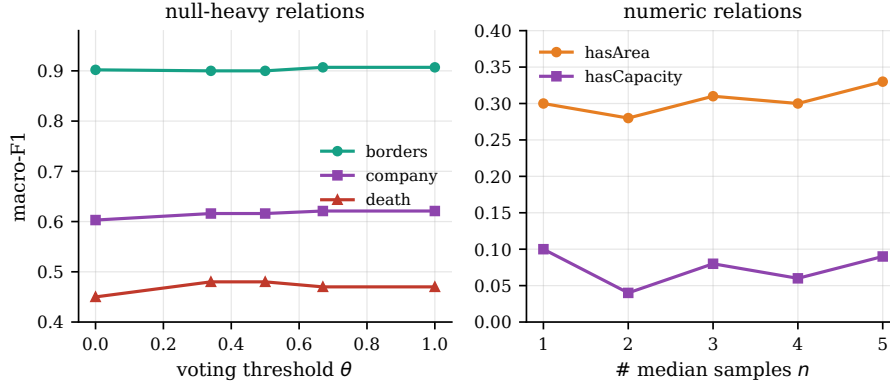


Figure 4: Left: per-relation  $F_1$  vs. voting threshold  $\theta$  (Qwen2.5-14B) for the three null-heavy relations (borders, company, death); the most null-heavy one (death, 45% empty) peaks at a lower  $\theta$  than the others. Right: numeric  $F_1$  vs. number of median samples  $n$  for HASAREA and HASCAPACITY; aggregation is roughly neutral, indicating a knowledge-bound gap.

| Relation | Subject           | Gold (abbrev.)                                     | Gemma-4-31B output                          |       |
|----------|-------------------|----------------------------------------------------|---------------------------------------------|-------|
| borders  | Ethiopia          | Djibouti, Eritrea, Kenya, Somalia, S. Sudan, Sudan | Djibouti, Eritrea, Kenya, Somalia, S. Sudan | ✓ 5/6 |
| death    | Vito Acconci      | New York City                                      | New York City                               | ✓     |
| company  | Bharti Airtel     | Bombay Stock Exchange                              | NSE of India; Bombay Stock Exchange         | ✓     |
| capacity | Gileno de Carli   | 5000                                               | 5000                                        | ✓     |
| area     | Lošinj            | 74.36                                              | 123.3                                       | ×     |
| award    | Stockholm hon. dr | Torvalds, Hedlund, Trebst, ... (40)                | Mandela, Tutu, Annan, Yunus, ...            | ×     |

Table 5: Qualitative examples: Gemma-4-31B vs. gold. Exact hits on PERSONHASCITY-OFDEATH/HASCAPACITY, near-complete borders; the numeric near-miss (HASAREA) and plausible-but-wrong enumeration (AWARDWONBY) are the knowledge-bound errors discussed above.

## Limitations

**Held-out evaluation.** We report validation results only; no test-set numbers are available yet. Our reported systems use a fixed, a-priori  $\theta=0.5$ , but the diagnostic threshold sweep (Figure 4) is run on the same split it is scored on, so its per-relation optima are optimistic rather than held-out and should be read as diagnostic only.

**Decoding variance.** We did not measure the run-to-run variance of the stochastic decoding ( $N=3$ ,  $n=5$ ), and AWARDWONBY has only 10 subjects. Several margins are within plausible decoding noise, so small deltas, such as the 0.02 gap between the 14B and 30B on PERSONHASCITY-OFDEATH, are indicative only and should not be read as reliable per-relation orderings.

**Model coverage.** Being closed-book, the system inherits the base model’s coverage and recency limits and cannot recover facts the model never stored. This caps AWARDWONBY recall and rare-venue numeric accuracy, and it means our gains reflect parametric recall rather than any external grounding.

**Single-sample 30B.** Because API rate limits precluded multi-sample decoding, our Gemma-4-31B scores are single-sample. We therefore cannot report how the 30B behaves under the self-consistency and abstention voting that help the smaller models; pairing them is left to future work and may further lift the null-heavy and numeric relations, so our 30B numbers are likely a floor rather than a ceiling.

**Metric specificity.** The abstention advantage stems from this task’s macro metric and its empty-as-valid-answer setting. It may not transfer to single-answer or micro-only regimes, and practitioners porting the method to other benchmarks should re-verify that abstention is still a positive-value action before adopting the voting machinery.

## Ethics and Generative-AI Declaration

Closed-book KB construction surfaces facts from model parameters that may be stale, biased, or hallucinated. Predictions about people (e.g. place of death) can be wrong, so they should not be treated as authoritative without verification. Our abstention mechanism partly mitigates this: when the

model is uncertain it emits an empty answer rather than a confident fabrication. The only generative-AI component is the open-weight LM that produces the predictions, as the task requires; we used no proprietary assistant to generate results.

## References

Dimitrios Alivanistos, Selene Báez Santamaría, Michael Cochez, Jan-Christoph Kalo, Emile van Krieken, and Thiviyan Thanapalasingam. 2022. Prompting as probing: Using language models for knowledge base construction. In *Semantic Web Challenge on KB Construction from Pre-trained Language Models (LM-KBC)*.

Badr AlKhamissi, Millicent Li, Asli Celikyilmaz, Mona Diab, and Marjan Ghazvininejad. 2022. A review on language models as knowledge bases. *arXiv preprint arXiv:2204.06031*.

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*.

Tom B. Brown and 1 others. 2020. Language models are few-shot learners. *NeurIPS*.

Boxi Cao, Hongyu Lin, Xianpei Han, Le Sun, Lingyong Yan, Meng Liao, Tong Xue, and Jin Xu. 2021. Knowledgeable or educated guess? revisiting language models as knowledge bases. In *ACL*.

Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. 2023. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*.

Jeremy R. Cole, Michael J.Q. Zhang, Daniel Gillick, Julian Martin Eisenschlos, Bhuwan Dhingra, and Jacob Eisenstein. 2023. Selectively answering ambiguous questions. In *EMNLP*.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *NeurIPS*.

Xin Dong, Evgeniy Gabilovich, Jeremy Heitz, Wilko Horn, Ni Lao, Kevin Murphy, Thomas Strohmman, Shaohua Sun, and Wei Zhang. 2014. Knowledge vault: A web-scale approach to probabilistic knowledge fusion. In *KDD*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *NeurIPS*.

Gemma Team. 2026. Gemma 4. Google DeepMind. Open-weight models under the Apache 2.0 license; we use the 31B dense variant. Model card: [https://ai.google.dev/gemma/docs/core/model\\_card\\_4](https://ai.google.dev/gemma/docs/core/model_card_4).

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *ICML*.

Kelvin Guu, Kenton Lee, Zora Tung, Panupong Papat, and Ming-Wei Chang. 2020. REALM: Retrieval-augmented language model pre-training. In *ICML*.

Zhengbao Jiang, Frank F. Xu, Jun Araki, and Graham Neubig. 2020. How can we know what language models know? *Transactions of the ACL*.

Saurav Kadavath, Tom Conerly, Amanda Askell, and 1 others. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Jan-Christoph Kalo, Tuan-Phong Nguyen, Simon Razniewski, and Bohui Zhang. 2024. Preface: Knowledge base construction from pre-trained language models (lm-kbc) 2024. In *LM-KBC Challenge at ISWC, CEUR Workshop Proceedings*.

Jan-Christoph Kalo, Simon Razniewski, Bohui Zhang, and Tuan-Phong Nguyen. 2025. LM-KBC 2025: 4th challenge on knowledge base construction from pre-trained language models. In *LM-KBC Challenge, CEUR Workshop Proceedings*.

Jan-Christoph Kalo, Sneha Singhania, Simon Razniewski, and Jeff Z. Pan. 2023. LM-KBC 2023: The 2nd challenge on knowledge base construction from pre-trained language models. In *Knowledge Base Construction from Pre-trained Language Models (LM-KBC), CEUR Workshop Proceedings*.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*.

Patrick Lewis and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*.

Alex Mullen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating the effectiveness of parametric and non-parametric memories. In *ACL*.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*.

Jeff Z. Pan, Simon Razniewski, Jan-Christoph Kalo, and 1 others. 2023. Large language models and knowledge graphs: Opportunities and challenges. *Transactions on Graph Data and Knowledge (TGDK)*.

Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases? In *EMNLP-IJCNLP*.



- Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. Know what you don’t know: Unanswerable questions for SQuAD. In *ACL*.
- Adam Roberts, Colin Raffel, and Noam Shazeer. 2020. How much knowledge can you pack into the parameters of a language model? In *EMNLP*.
- Taylor Shin, Yasaman Razeghi, Robert L. Logan IV, Eric Wallace, and Sameer Singh. 2020. AutoPrompt: Eliciting knowledge from language models with automatically generated prompts. In *EMNLP*.
- Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10).
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *ICLR*.
- Bohui Zhang, Ioannis Reklos, Nitisha Jain, Albert Meroño-Peñuela, and Elena Simperl. 2023. Using large language models for knowledge engineering (LLMKE): A case study on wikidata. *arXiv preprint arXiv:2309.08491*.
- Zexuan Zhong, Dan Friedman, and Danqi Chen. 2021. Factual probing is [MASK]: Learning vs. learning to recall. In *NAACL*.